

Artificial Neural Networks

Q6. Compute Perceptron Output

Inputs: $x_1 = 3$, $x_2 = 1$

Weights: $w_1 = 0.4$, $w_2 = -0.3$

Bias = 0.2

Activation: Linear

Solution:

$$z = 0.4(3) + (-0.3)(1) + 0.2 = 1.1$$

Q7. Sigmoid Neuron Output

Inputs: $x_1 = 2$, $x_2 = 5$

Weights: $w_1 = 0.1$, $w_2 = 0.4$

Bias = -0.6

Solution:

$$z = 0.1(2) + 0.4(5) - 0.6 = 2.0$$

$$\text{Output} = 1/(1+e^{-2}) \approx 0.881$$

Q8. ReLU Activation

Inputs: $x_1 = -3$, $x_2 = 4$

Weights: $w_1 = -0.2$, $w_2 = 0.7$

Bias = 1

Solution:

$$z = -0.2(-3) + 0.7(4) + 1 = 4.6$$

$$\text{ReLU Output} = 4.6$$

Q9. Two-Layer ANN

Hidden layer neuron:

Weights = (0.3, 0.9), Bias = 0.4

Input = (1, 3)

Output layer weight = 0.5, bias = -0.2

Solution:

Hidden $z = 0.3(1) + 0.9(3) + 0.4 = 3.4$

Hidden Sigmoid ≈ 0.967

Final Output = $0.5(0.967) - 0.2 = 0.2835$